DAA PACKAGE ABSTRACT

PACKAGE TITLE: **DNA MOTIF DISCOVERY AND STATISTICAL PROFILING**

The DNA Motif Discovery and Statistical Profiling system is a Design and Analysis of Algorithms (DAA) project designed to demonstrate the efficient identification and analysis of exact nucleotide patterns in genomic sequences using the linear-time Z-Algorithm. The project focuses on discovering biologically significant DNA motifs across real genome samples and evaluating their frequency distributions to establish their role in molecular biology and disease diagnosis.

The application models the linear-time string matching process by constructing a concatenated string of the motif pattern, a unique sentinel delimiter, and the genomic text. It illustrates the fundamental working of the Z-Algorithm, which computes the length of the longest substring starting at each position that matches a prefix of the string in $O(n + m)$ time. By maintaining an active matching window defined by [L, R] boundaries, the algorithm reuses previously computed prefix matches to avoid redundant character comparisons, dramatically outperforming naive search methods on highly repetitive genomic sequences.

The project aims to provide an interactive and visual representation of motif occurrences and statistical profiles across diverse biological sequences sourced from NCBI GenBank. It incorporates a Markov null model calibrated to background nucleotide (GC) content to compute statistical enrichment Z-scores and identify significant tandem repeat expansions. Users can observe algorithmic telemetry, motif distributions, and comparative profiles across different specimens to classify samples into normal, carrier, and pathogenic risk categories.

This project helps develop a practical understanding of advanced linear-time string matching algorithms, algorithm efficiency optimization, and statistical data modeling in computational biology. It can serve as an educational and analytical tool for understanding how fundamental algorithmic techniques like the Z-Algorithm can be leveraged to analyze massive genomic datasets and predict genetic disorders reliably.

TEAM MEMBERS

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